

1. Which command installs a new package in R?

- (a) `require()`
- (b) `library()`
- (c) `install.packages()`
- (d) `load()`

2. What does the following code return?

```
gss24 |>
  group_by(sex) |>
  summarize(mean_age = mean(age, na.rm = TRUE))
```

- (a) a data frame showing the average age for each sex.
- (b) A filtered data frame with only rows where age is not missing.
- (c) A new column named `mean_age` added to the original data frame.
- (d) a list of all ages grouped by sex.

3. What is the result of the following code?

```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```

- (a) rows with `age > 20` are removed
- (b) The original `gss24` dataset is overwritten.
- (c) Only rows with `age > 20` and columns `age` and `childs` are returned.
- (d) A new column is added to `gss24`

4. The variable `mntlhlth` measures how many days in the past 30 a respondent reported poor mental health. A respondent reports 5 poor mental health days. Based on the summary statistics for this variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.53	7.93

- (a) The z-score is negative because 5 is below the sd.
- (b) The z-score is close to zero because 5 is close to the mean.
- (c) The z-score is undefined because the median is 0.
- (d) The z-score is large and positive because 5 is much higher than the mean.

5. If a statistics classes' final grades were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?

- (a) 50%
- (b) 62%
- (c) 95%
- (d) 68%

6. Which RStudio pane is used to write and edit R scripts and Quarto documents?

- (a) Source
- (b) Console
- (c) Environment
- (d) Files

7. Which function is used to load a package in R?
- (a) `load()`
 - (b) `install.packages()`
 - (c) `library()`
 - (d) `source()`
8. What does the `::` operator do in R?
- (a) Imports all functions from a package
 - (b) Assigns a value to a variable
 - (c) Specifies a function from a particular package (without loading it)
 - (d) Comments out a line of code
9. What will happen if you try to use a function from a package that hasn't been loaded?
- (a) The function will run normally
 - (b) You will get an error saying the function is not found
 - (c) R will automatically load the package
 - (d) R will prompt you to install the package

10. The level of measurement for the `hapmar` variable is _____

[1] "Happiness of marriage" Frequencies

`gss24$hapmar`

Type: Factor

	Freq	%	% Cum.
very happy	802	59.36	59.36
pretty happy	504	37.31	96.67
not too happy	45	3.33	100.00
Total	1351	100.00	100.00

- (a) interval
 - (b) dichotomous
 - (c) nominal
 - (d) ordinal
11. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?
- (a) This couple had fewer arguments than about 5% of couples.
 - (b) This couple had about 8 arguments per month.
 - (c) This couple had fewer arguments than about 2% of couples.
 - (d) This couple had fewer arguments than about 98% of couples.
12. In a normally distributed set of test scores, a student scoring below the average will have a _____.
- (a) positive z-score
 - (b) negative standard error
 - (c) positive standard error
 - (d) negative z-score
13. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?
- (a) frequencies
 - (b) cumulative frequencies
 - (c) percentages
 - (d) cumulative percentages

14. Which of the following is a measure of the spread of a distribution?
- (a) median
 - (b) standardized variable
 - (c) mean
 - (d) standard deviation
15. In *Naked Statistics*, Charles Wheelan explains that “most sample means will lie reasonably close to the population mean; the standard error is what defines ‘reasonably close.’” What does this statement imply?
- (a) The standard error increases as sample size increases.
 - (b) The standard error tells us how far the population mean is from the sample mean.
 - (c) The standard error measures how much sample means tend to vary around the population mean.
 - (d) The sample mean is always equal to the population mean.
16. What does the pipe operator (`|>`) do in `dplyr`?
- (a) It creates a new column in a data frame.
 - (b) It passes the result of one function into the next.
 - (c) It filters rows based on a condition.
 - (d) It loads the `dplyr` package into memory.
17. Which of the following is a valid use of `group_by()` and `summarize()`?
- (a) `group_by(data, summarize(sex))`
 - (b) `summarize(group_by(data, age), mean(sex))`
 - (c) `data |> summarize(mean = group_by(sex))`
 - (d) `data |> group_by(sex) |> summarize(freq = n())`
18. The U.S. General Social Survey asked people how many brothers and sisters they had. A researcher used R to produce the summary statistics of this variable (`sibs`) below. Based on this, you conclude the distribution for this variable is:
- | freq | median | mean | sd |
|------|--------|------|------|
| 3291 | 3 | 2.92 | 1.84 |
- (a) nominal
 - (b) negatively skewed
 - (c) symmetrical
 - (d) positively skewed
19. What is Wheelan’s primary message of Chapter 1 (“What’s the Point?”) in *Naked Statistics: Stripping the Dread from the Data*?
- (a) Statistics simplify complex information to help us make better decisions
 - (b) Statistical formulas are the most important part of understanding data
 - (c) Statistics provide honest answers to complex questions
 - (d) Statistics explain why relationships exist between variables
20. What does Oscar Gandy suggest about the use of racial statistics?
- (a) Their use in data journalism helps mitigate online polarization
 - (b) They can both expose and perpetuate discrimination
 - (c) They are neutral tools most useful for academic research
 - (d) Their use in legal proceedings increasingly lead to unfair policy outcomes

21. Which of these datasets has a standard deviation of 0?
- (a) 0, 2, 4, 6, 8
 - (b) 4, 4, 4, 4, 4
 - (c) 2, 2, 5, 7, 7
 - (d) 1, 4, 7, 10, 15
22. Which of the following code snippets correctly uses `mutate()` to convert height from centimeters to meters?
- (a) `mutate(height <- height / 100)`
 - (b) `mutate(height = select(height / 100))`
 - (c) `mutate(height / 100 = height)`
 - (d) `mutate(height = height / 100)`
23. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?
- (a) Their score is unusual and higher than most.
 - (b) Their score is very common.
 - (c) Their score is within the middle 68% of the distribution.
 - (d) Their score is likely below average.
24. In order to compare meaningfully across groups, a frequency should be converted into a/an _____.
- (a) number
 - (b) standard deviation
 - (c) ordinal variable
 - (d) percentage
25. Based on the table below, which statement best compares the relative frequency of being `not too happy` between men and women?

Cross-Tabulation, Column Proportions

`happy * sex`

Data Frame: `gss24`

	sex	male	female	Total
happy				
very happy		293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy		830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy		330 (22.7%)	370 (20.4%)	700 (21.5%)
Total		1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

- (a) The frequency of being `not too happy` is equal for men and women.
 - (b) Women have a slightly higher relative frequency because more women than men reported being `not too happy`.
 - (c) Men have a slightly higher relative frequency of being `not too happy` than women, even though fewer men reported it.
 - (d) A greater number of women reported being `not too happy` than men, so women have a higher relative frequency.
26. What is the purpose of `drop_na()` in `dplyr`?
- (a) It converts NA values to NULL.
 - (b) It drops columns that are entirely NA.
 - (c) It removes rows that contain missing values.
 - (d) It replaces missing values with zeros.

27. What does the expression `5 != 3` evaluate to in R?
- (a) TRUE
 - (b) FALSE
 - (c) NULL
 - (d) NA

1. (a) ☐ (b) ☐ (c) ☒ (d) ☐
2. (a) ☒ (b) ☐ (c) ☐ (d) ☐
3. (a) ☐ (b) ☐ (c) ☒ (d) ☐
4. (a) ☐ (b) ☒ (c) ☐ (d) ☐
5. (a) ☐ (b) ☐ (c) ☐ (d) ☒
6. (a) ☒ (b) ☐ (c) ☐ (d) ☐
7. (a) ☐ (b) ☐ (c) ☒ (d) ☐
8. (a) ☐ (b) ☐ (c) ☒ (d) ☐
9. (a) ☐ (b) ☒ (c) ☐ (d) ☐
10. (a) ☐ (b) ☐ (c) ☐ (d) ☒
11. (a) ☐ (b) ☐ (c) ☐ (d) ☒
12. (a) ☐ (b) ☐ (c) ☐ (d) ☒
13. (a) ☐ (b) ☐ (c) ☐ (d) ☒
14. (a) ☐ (b) ☐ (c) ☐ (d) ☒
15. (a) ☐ (b) ☐ (c) ☒ (d) ☐
16. (a) ☐ (b) ☒ (c) ☐ (d) ☐
17. (a) ☐ (b) ☐ (c) ☐ (d) ☒
18. (a) ☐ (b) ☒ (c) ☐ (d) ☐
19. (a) ☒ (b) ☐ (c) ☐ (d) ☐
20. (a) ☐ (b) ☒ (c) ☐ (d) ☐
21. (a) ☐ (b) ☒ (c) ☐ (d) ☐
22. (a) ☐ (b) ☐ (c) ☐ (d) ☒

23. (a) ☒ (b) ☐ (c) ☐ (d) ☐
24. (a) ☐ (b) ☐ (c) ☐ (d) ☒
25. (a) ☐ (b) ☐ (c) ☒ (d) ☐
26. (a) ☐ (b) ☐ (c) ☒ (d) ☐
27. (a) ☒ (b) ☐ (c) ☐ (d) ☐